THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

Dec 16, 2017 TIME: 9:30 am – 12:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg.°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. An oligotrophic lake is typically (a) high in N concentration, (b) high in P concentration, (c) high in DO concentration, (d) high in chlorophyll concentration, and (e) high in turbidity.
2. The BOD of a wastewater is estimated to be around 300 mg/L. What is the most suitable dilution ratio for performing a BOD test? (a) no dilution needed, (b) 1:10 dilution, (c) 1:100 dilution, (d) 1:1000 dilution, or (e) 1: 10000 dilution.
3. Which of the following water treatment process is NOT intended for particle removal? (a) primary sedimentation, (b) secondary sedimentation, (c) coagulation, (d) flocculation, (e) softening, (f) filtration, or (g) none of the above.
4. The F/M ratio is an important parameter in activated sludge process. M is typically represented by (a) TDS, (b) TSS, (c) TVS, (d) TFS, (e) FSS, (f) VSS, (g) FDS, or (h) none of the above.
5. Which of the following mechanisms is NOT important in coagulation process? (a) Charge neutralization, (b) entrapment, (c) charge screening, (d) inter-particle bridging, or (e) charge reversal.
6. Which biological treatment process does not require oxygen? (a) Activated sludge, (b) aerated lagoon, (c) trickling filter, (d) rotating biological contactor, or (e) none of the above.
7. Which membrane type is suitable for MBR? (a) RO, (b) NF, (c) UF, (d) depth filter, (e) multimedia filter, or (f) none of the above.
8. How does dissolving CO_2 into a water solution change its alkalinity (ALK)? (a) ALK increases, (b) ALK does not change, (c) ALK decreases, or (d) ALK decreases initially and then increases.
9. Which process is typically used for NO_x removal from vehicle emission? (a) Fuel cleaning, (b) three-way catalytic convertor, (c) scrubber, (d) selective catalytic reduction, (e) cyclone, or (f) none of the above.
10. The radioactive decay of strontium-90 is given by ${}^{90}_{38}Sr \rightarrow {}^{90}_{39}Y + x$. The particle x is best described as (a) α particle, (b) β particle, (c) photon, or (d) a particle of unknown identity.

Section B (Three questions with unequal marks)

11. (a) Briefly explain the following terminologies (**use < 80 words for each item**):
    i. Acid rain;
    ii. Ocean acidification;
    iii. Secondary disinfectant;
    iv. Disinfection byproducts.

    (20 marks)

    (b) Some lakes and reservoirs experience thermal stratification in winters. Briefly explain this phenomenon by sketching the temperature profile in a winter stratified lake. Do you expect winter stratification to occur in reservoirs in Hong Kong?

    (10 marks)

12. (a) A wastewater contains 200 mg/L biodegradable organic chemical (represented by $C_5H_{10}O_5$). Determine its theoretical oxygen demand and nitrogenous oxygen demand.

(6 marks)

(b) This wastewater is accidentally discharged to a nearby river. The contaminated river water has an organic concentration of 25 mg/L and a temperature of 20 °C. The flow velocity of river is 1 km/h. A city is located along the river at 50 km downstream to the discharge point. Estimate the concentration of organic chemical remaining in the river water by the time it reaches to the city. State the type of reactor model you used. You can assume a biodegradation rate constant of 0.20 day^{-1} at 25°C and a temperature correction factor of 1.047 in your calculation(s).

(9 marks)

13. A water solution (at 25°C) has a pH of 7.3, and it contains 40 mg/L calcium (Ca^{2+}) and 12 mg/L magnesium (Mg^{2+}).

(a) Determine $[H_2CO_3]$, $[HCO_3^-]$, and $[CO_3^{2-}]$. Give your answers in molar concentration (M).

(9 marks)

(b) Determine the alkalinity, total hardness, carbonate hardness, and non-carbonate hardness. Express your answers in mg/L as $CaCO_3$ equivalent.

(10 marks)

(c) Check the scaling potential of the solution.

(4 marks)

(d) Recommend a method for hardness removal.

(2 marks)

Hints:

i. You can assume the water has reached chemical equilibrium with the atmosphere (390 ppm CO_2 in atmosphere).

ii. You may need to use the following thermodynamic data in your calculations:
- Henry's constant for CO_2 at 25°C: $K_H = 3.39 \times 10^{-2}$ M/atm
- Acidity constants of carbonic acid (H_2CO_3) at 25°C: $pK_1 = 6.3$; $pK_2 = 10.3$
- Solubility of $CaCO_3$(s): $pK_{so} = 8.3$

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